

THE WEAKNESS OF TYPICALITY

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ABSTRACT. Many statements studied in reverse mathematics can be seen as mathematical problems, formulated in terms of instances and solutions. We develop a framework of typicality encompassing measure and genericity, and we classify the reverse mathematics zoo in terms of which problems admit typical solutions. It turns out that even very weak problems do not admit typical solutions.

1. INTRODUCTION

In mathematical practice, it is common to define objects as “typical” if they avoid all extraordinary circumstances up to a certain degree of complexity. For instance, a set of points is in general linear position if it satisfies no more linear relations than are necessary; such a set might be said to be typical with respect to linear relations.

In Cantor space (2^ω), two notions of typicality currently dominate: Martin-Löf n -randomness, or avoidance of all “effectively null” classes as defined by Σ_n^0 Martin-Löf tests; and weak Cohen n -genericity, or avoidance of all meager classes whose complements are countable intersections of classes described by Σ_n^0 sets of strings.

We know that each sort of typicality forces certain properties for subsets of ω . For example, no n -generic or n -random set is computable (or in fact Σ_n^0 , for obvious reasons), but no 1-generic computes a DNC function; on the other hand, every 1-random does compute a DNC function. In fact, increasing the level of randomness can significantly increase the power of a typical oracle. The Rainbow Ramsey Theorem for pairs and 2-bounded colors states that every 2-bounded coloring of pairs in ω has a “rainbow”, an infinite subdomain on which no color appears twice. There are known computable 2-bounded colorings with no computable rainbow; however, as shown by Csima and Mileti [7], every 2-random oracle computes rainbows for every computable 2-bounded coloring.

Applying (the relativized versions of) these results in reverse mathematics, we see that axioms guaranteeing the mere existence of sufficiently typical sets can have useful implications:

Theorem 1.1. $\text{RCA}_0 + 1\text{-RAN} \vdash \text{DNC}$.

Theorem 1.2. $\text{RCA}_0 + 2\text{-RAN} \vdash \text{RRT}_2^2$.

This raises further questions. Is there a useful consequence of 3-RAN? What about principles guaranteeing the existence of sufficiently generic sets? How far can these implications go, and are there any results that cannot be proven using

typicality alone? Fundamentally, we ask: is the existence of strongly typical sets a weak principle, or a strong assumption?

As suggested by our title, we find that typicality principles are weak. Specifically, we establish strict upper bounds on the set of results implied by randomness- and genericity-existence principles.

In Section 2, we begin by constructing a broad theorem, applicable to all notions of typicality in a broad framework. This framework is fairly simple, and is satisfied by both standard typicality notions on 2^ω . Our theorem is designed for implications between problems and more precisely Π_2^1 -sentences, but our analysis in Section 3 also offers a variation for more general second-order statements.

In Sections 4 and 5, we explore applications to randomness- and genericity-existence principles respectively. We successfully classify most standard reverse-mathematical principles as either consequences of randomness, or not provable from any randomness-existence principle. We conduct a similar analysis for genericity.

A note to the reader: in this paper we refer to subsets of 2^ω as *classes*, in order to distinguish them from points in 2^ω , which we think of as *sets*.

2. FRAMEWORK

This section is devoted to the development of an abstract framework for studying the existence of typical solutions to mathematical problems. In Section 2.1, we give an axiomatic definition of smallness, encompassing randomness and genericity. In Section 2.2, we define the abstract notion of mathematical problem, in terms of instances and solutions, and compare them with respect to computable and ω -reducibilities. In Section 2.3, we define what it means for a problem to be frequently solved, and prove our main theorem, stating that being frequently solved is upward-closed under ω -reducibility. Last, in Section 3, we refine our notion of frequently solved problem to obtain three properties, and obtain further separations.

2.1. Smallness and Typicality. Suppose we have a property of classes $\mathcal{C} \subseteq 2^\omega$, called *smallness*, with the following properties:

- (1) $\emptyset$ is small, and 2^ω is not.
- (2) If $\mathcal{B}$ is small and $\mathcal{A} \subseteq \mathcal{B}$, then $\mathcal{A}$ is small.
- (3) If $\mathcal{B}_i$ is small for all $i \in \omega$, then $\bigcup_{i \in \omega} \mathcal{B}_i$ is also small.
- (4) Smallness is closed under the operation:

$$\mathcal{B}(\mathcal{S}) = \{X : \{Z : X \oplus Z \in \mathcal{S}\} \text{ is not small}\};$$

that is, if $\mathcal{S}$ is small, so is $\mathcal{B}(\mathcal{S})$.

Properties (1)-(3) are those of a proper σ -ideal; thus, for most purposes we merely add that our notion of smallness satisfies property (4).

We can define a class to be small (in the sense of measure) if it has measure 0; property (4) then holds as a consequence of Fubini's Theorem. In this case, a small class $\mathcal{C}$ is said to be n -small if it has a Σ_n^0 -Martin-Löf test; that is, if there is a sequence of uniformly Σ_n^0 classes $\{W_n\}$ such that $\mu(W_n) \leq 2^{-n}$ for all n , with $\mathcal{C} \subseteq \bigcap_{n < \omega} W_n$. More generally, $\mathcal{C}$ is n - X -small if we replace Σ_n^0 by $\Sigma_n^{0,X}$ in this definition.

We can also define a class to be small (in the sense of category) if it is meager; property (4) is then essentially equivalent to the weak direction of the Kuratowski-Ulam Theorem. In this case, we say that a small class $\mathcal{C}$ is n -small if its complement

is covered by an intersection of classes described by dense Σ_n^0 sets of strings. More generally, C is n - X -small if we replace Σ_n^0 by $\Sigma_n^{0,X}$ in this definition.

A set is *typical* if it avoids all small classes up to some descriptive complexity. Specifically, $T \subseteq \omega$ is n - X -typical if it avoids all n - X -small classes. We note that the class of all n - X -typical sets, as a countable intersection of co-small classes, will always be co-small; in particular, for all n and all X , every non-small class has an n - X -typical member.

2.2. Problems and reducibilities. A *problem* is a relation $P \subseteq 2^\omega \times 2^\omega$. A *P-instance* is any element of $\text{dom } P = \{X : \exists Y (X, Y) \in P\}$, and a *P-solution* to X is any element of $P(X) = \{Y : (X, Y) \in P\}$. Many statements studied in reverse mathematics are of the form

$$(\forall X)[\phi(X) \rightarrow (\exists Y)[\psi(X, Y)]] ,$$

where $\phi(X)$ and $\psi(X, Y)$ are arithmetic formulas. Any such statement can be seen as a problem P where $\text{dom } P = \{X : \phi(X)\}$ and $P(X) = \{Y : \psi(X, Y)\}$. We call them Π_2^1 -problems. There exist multiple computability-theoretic notions of reducibility to compare the strength of problems. The arguably most natural one is computable reducibility:

Definition 2.1. A problem P is *computably reducible* to Q (written $P \leq_c Q$) if for every P -instance X , there is an X -computable Q -instance $\hat{X}$ such that for every Q -solution $\hat{Y}$ to $\hat{X}$, $\hat{Y} \oplus X$ -computes a P -solution to X .

The notion of computable reducibility was formally introduced by Dzhafarov [9], although already present in spirit in many prior works. A more general notion, which is the computability-theoretic counterpart of implication over RCA_0 , is ω -reducibility, formulated in terms of Turing ideals:

Definition 2.2.

- (1) A *Turing ideal* $\mathcal{M}$ is a non-empty collection of sets which is downward-closed under Turing reducibility ($\forall X \in \mathcal{M} \forall Y \leq_T X, Y \in \mathcal{M}$) and closed under the effective join ($\forall X, Y \in \mathcal{M}, X \oplus Y \in \mathcal{M}$).
- (2) A problem P *holds* in a Turing ideal $\mathcal{M}$ (written $\mathcal{M} \models P$) if for every P -instance $X \in \mathcal{M}$, there is a P -solution $Y \in \mathcal{M}$.
- (3) A problem P is ω -*reducible* to Q (written $P \leq_\omega Q$) if for every Turing ideal $\mathcal{M}$, if $\mathcal{M} \models Q$ then $\mathcal{M} \models P$.

Note that ω -reducibility is coarser than computable reducibility, in that if $P \leq_c Q$, then $P \leq_\omega Q$, but the converse does not hold in general. Also note that if P and Q are Π_2^1 -problems such that $\text{RCA}_0 \vdash Q \rightarrow P$, then $P \leq_\omega Q$. Most separations in reverse mathematics are actually proofs of non-reduction for ω -reducibility. When restricted to Π_2^1 -problems, ω -reducibility is a particular case of the notion of computable entailment introduced by Shore [26].

2.3. Main Theorem.

Definition 2.3. A problem Q is *frequently solved* if, for every Q -instance X ,

$$\{Z : (\exists Y \leq_T X \oplus Z)[Y \text{ is a } Q\text{-solution to } X]\} \text{ is not small;}$$

that is, each instance has solutions below a non-small class of oracles, and thus below some typical oracle at any complexity.

In the sense of category, our prototypical examples of frequently solved principles have the form “For every X , there exists a sufficiently X -generic set”; in the sense of measure, we use “For every X , there exists a sufficiently X -random set”.

By contrast, fixing a problem P , we say that

Definition 2.4. A P -instance X is *rarely solved* if

$$\{Z : (\exists Y \leq_T X \oplus Z)[Y \text{ is a } P\text{-solution to } X]\} \text{ is small;}$$

that is, X has solutions below only a small class of oracles (even when also provided the instance X). In particular, no sufficiently typical oracle can solve X .

Naturally, a principle has a rarely-solved instance if and only if it is not frequently solved. For our main theorem, we show that being frequently solved is preserved by ω -reducibility. Specifically,

Theorem 2.5. *If P is frequently solved, but Q has a rarely-solved instance X , then $Q \not\leq_\omega P$.*

In particular, if P and Q are Π_2^1 -problems such that P is frequently solved, but Q has a rarely-solved instance X , then P does not imply Q over RCA_0 .

To construct the Turing ideal witnessing $Q \not\leq_\omega P$, we will begin with a Turing ideal containing only sets computable from X ; since X is rarely solved, this includes no solutions to the Q -instance. We then add solutions to all P -instances while avoiding $\mathcal{C}$, the class of all oracles that solve X . However, we need to prevent added P -solutions from *jointly* forcing us to add a Q -solution to X ; we do this by instead avoiding any set with respect to which X could become frequently solved. This requires we start with some technical lemmas; we first show that the $\mathcal{B}$ -operator from smallness axiom (4) expands $\mathcal{C}$ to include more of these dangers, and then show that iterating this expansion tends towards a fixed point that remains small.

Definition 2.6. Let $\mathcal{C} \subseteq 2^\omega$. We say $\mathcal{C}$ is an X_0 -computational class if for any $X \in \mathcal{C}$ and any Y with $X_0 \oplus Y \geq_T X$, $Y \in \mathcal{C}$; that is, if $\mathcal{C}$ is closed upwards under Turing reducibility relative to X_0 .

The following simple lemma states an important property of small X_0 -computational classes: they avoid X_0 . As we shall expand our small class by iterating the $\mathcal{B}$ -operator, proving that the operator preserves being small and X_0 -computational is a way to ensure that the expanded class still does not contain X_0 .

Lemma 2.7. *If $\mathcal{C}$ is a small X_0 -computational class, then $X_0 \notin \mathcal{C}$.*

Proof. Suppose for the contradiction that $X_0 \in \mathcal{C}$. Then, since $X_0 \oplus Z \geq_T X_0$ for any set $Z \in 2^\omega$, $Z \in \mathcal{C}$, so $\mathcal{C} = 2^\omega$ which is not small. $\square$

The next lemma shows that the $\mathcal{B}$ -operator preserves being small and an X_0 -computational class:

Lemma 2.8. *If $\mathcal{S} \subseteq 2^\omega$ is a small X_0 -computational class, then*

$$\begin{aligned} \mathcal{B}(\mathcal{S}) &= \{X : \{Z : X_0 \oplus X \oplus Z \in \mathcal{S}\} \text{ is not small}\} \\ &= \{X : \{Z : X \oplus Z \in \mathcal{S}\} \text{ is not small}\}. \end{aligned}$$

is a small X_0 -computational class with $\mathcal{S} \subseteq \mathcal{B}(\mathcal{S})$.

Proof. First, we note that the two specifications of $\mathcal{B}(\mathcal{S})$ are in fact the same, as $X_0 \oplus X \oplus Z \in \mathcal{S}$ if and only if $X \oplus Z \in \mathcal{S}$. (After all, $X_0 \oplus X \oplus Z$ is equivalent to $X \oplus Z$ under Turing reducibility relative to X_0 .)

Consider $X \in \mathcal{S}$. Since $X \oplus Z \geq_T X$ and $\mathcal{S}$ is a X_0 -computational class, $X \oplus Z \in \mathcal{S}$ for every $Z \in 2^\omega$. Thus, $\{Z : X \oplus Z \in \mathcal{S}\} = 2^\omega$ is certainly not small, so $X \in \mathcal{B}(\mathcal{S})$. Since X was arbitrary, $\mathcal{S} \subseteq \mathcal{B}(\mathcal{S})$.

Next, suppose $X \in \mathcal{B}(\mathcal{S})$ and $X_0 \oplus Y \geq_T X$. Clearly, $X_0 \oplus Y \oplus Z \geq_T X \oplus Z$ for any $Z \in 2^\omega$. Thus, since $\mathcal{S}$ is an X_0 -computational class, $X \oplus Z \in \mathcal{S}$ implies $Y \oplus Z \in \mathcal{S}$. This means that $\{Z : Y \oplus Z \in \mathcal{S}\} \supseteq \{Z : X \oplus Z \in \mathcal{S}\}$ (which is not small), so $Y \in \mathcal{B}(\mathcal{S})$. Therefore, $\mathcal{B}(\mathcal{S})$ is an X_0 -computational class.

Last, since $\mathcal{S}$ is small, $\mathcal{B}(\mathcal{S})$ is small by property (4). $\square$

Given a small X_0 -computational class $\mathcal{C}$, let $\mathcal{B}_\mathcal{C}^0 = \mathcal{C}$, and define $\mathcal{B}_\mathcal{C}^{n+1} = \mathcal{B}(\mathcal{B}_\mathcal{C}^n)$. By Lemma 2.8, these classes form an increasing chain of small X_0 -computational classes

$$\mathcal{C} = \mathcal{B}_\mathcal{C}^0 \subseteq \mathcal{B}_\mathcal{C}^1 \subseteq \mathcal{B}_\mathcal{C}^2 \subseteq \dots$$

Let $\mathcal{B}_\mathcal{C}^\omega = \bigcup_{n \in \omega} \mathcal{B}_\mathcal{C}^n$. We show that $\mathcal{B}_\mathcal{C}^\omega$ is a small X_0 -computational class and a fixed point for the $\mathcal{B}$ operator.

Lemma 2.9. *If $\mathcal{C}$ is a small X_0 -computational class, so is $\mathcal{B}_\mathcal{C}^\omega$.*

Proof. Since $\mathcal{B}_\mathcal{C}^\omega$ is an increasing countable union of small classes, it must itself be small. Let $X_0 \oplus Y \geq_T X$ for some $X \in \mathcal{B}_\mathcal{C}^\omega$. Since $X \in \mathcal{B}_\mathcal{C}^n$ for some n , and $\mathcal{B}_\mathcal{C}^n$ is an X_0 -computational class, $Y \in \mathcal{B}_\mathcal{C}^n \subseteq \mathcal{B}_\mathcal{C}^\omega$. Therefore, $\mathcal{B}_\mathcal{C}^\omega$ is an X_0 -computational class. $\square$

Lemma 2.10. *For any small X_0 -computational class $\mathcal{C}$, $\mathcal{B}(\mathcal{B}_\mathcal{C}^\omega) = \mathcal{B}_\mathcal{C}^\omega$.*

Proof. Suppose $X \in \mathcal{B}(\mathcal{B}_\mathcal{C}^\omega)$; that is, $\{Z : X_0 \oplus X \oplus Z \in \mathcal{B}_\mathcal{C}^\omega\}$ is not small. Let $\chi_n = \{Z : X_0 \oplus X \oplus Z \in \mathcal{B}_\mathcal{C}^n\}$; clearly, $\chi_n \subseteq \chi_{n+1}$, and

$$\bigcup_{n \in \omega} \chi_n = \{Z : X_0 \oplus X \oplus Z \in \mathcal{B}_\mathcal{C}^\omega\}.$$

Since this increasing countable union is not small, there must be some χ_n that is not small, so $X \in \mathcal{B}(\mathcal{B}_\mathcal{C}^n) = \mathcal{B}_\mathcal{C}^{n+1} \subseteq \mathcal{B}_\mathcal{C}^\omega$. We have thus established that $\mathcal{B}(\mathcal{B}_\mathcal{C}^\omega) \subseteq \mathcal{B}_\mathcal{C}^\omega$. The reverse inclusion follows directly from Lemma 2.9 and Lemma 2.8. $\square$

Towards our main theorem, we next show that for any P-instance, we can always add a solution without adding any elements of a small class.

Lemma 2.11. *Let P be a frequently solved problem, and let $\mathcal{C}$ be a small X_0 -computational class. Fix some $A \subseteq \omega$ such that $A \notin \mathcal{B}_\mathcal{C}^\omega$.*

If $X \leq_T A$ is a P-instance, then X has a P-solution Y such that $A \oplus Y \notin \mathcal{B}_\mathcal{C}^\omega$.

Proof. By Lemma 2.10, $A \notin \mathcal{B}(\mathcal{B}_\mathcal{C}^\omega)$. Thus, $\{Z : X_0 \oplus A \oplus Z \in \mathcal{B}_\mathcal{C}^\omega\}$ is small.

Since P is frequently solved, the class

$$\{Z : X \oplus Z \text{ computes a P-solution to } X\}$$

is not small. Therefore, there exist sets Z and Y such that $X \oplus Z \geq_T Y$, Y is a P-solution to X , and $X_0 \oplus A \oplus Z \notin \mathcal{B}_\mathcal{C}^\omega$. Since $A \geq_T X$ and $\mathcal{B}_\mathcal{C}^\omega$ is an X_0 -computational class, $A \oplus Y \notin \mathcal{B}_\mathcal{C}^\omega$. $\square$

Theorem 2.12. *Let P be a frequently-solved problem, and let $\mathcal{C}$ be a small X_0 -computational class. There exists a Turing ideal $\mathcal{M} \models P$ that contains X_0 , but includes no sets from $\mathcal{C}$.*

Proof. We start with the Turing ideal containing only the sets computable from X_0 . By Lemma 2.7 and Lemma 2.9, $X_0 \notin \mathcal{B}_{\mathcal{C}}^\omega$. Iterating Lemma 2.11, we add solutions to P -instances in the standard dovetailing manner while avoiding all computations of elements of $\mathcal{B}_{\mathcal{C}}^\omega$. Since $\mathcal{C} \subseteq \mathcal{B}_{\mathcal{C}}^\omega$, the resulting Turing ideal $\mathcal{M}$ will satisfy $\mathcal{M} \cap \mathcal{C} = \emptyset$. $\square$

Finally, we apply this to the class of oracles that solve a rarely-solved Q -instance, and obtain our main result.

Proof of Theorem 2.5. Fix some rarely-solved instance X_0 of Q , and let

$$\mathcal{C} = \{Z : X_0 \oplus Z \text{ computes a } Q\text{-solution to } X_0\}.$$

$\mathcal{C}$ is an X_0 -computational class by definition, and it is small by assumption. By Theorem 2.12, there is a Turing ideal $\mathcal{M}$ such that $\mathcal{M} \models P$, $\mathcal{M}$ includes X_0 , and $\mathcal{M}$ includes no sets from $\mathcal{C}$. In particular, $\mathcal{M}$ contains no Q -solutions to X_0 , so $\mathcal{M} \not\models Q$. $\square$

Remark 2.13. Note that the proof of Theorem 2.5 used the axiom (3) of smallness only for increasing sequences of small classes ($\mathcal{B}_0 \subseteq \mathcal{B}_1 \subseteq \dots$). One could therefore restrict the axiom to such sequences. However, as soon as we require our notion of smallness to be an ideal – which is a reasonable assumption removing pathological behaviors – the axiom of increasing countable unions implies the axiom of arbitrary countable unions.

3. ANALYSIS AND VARIANTS

On inspection, our notions of frequently-solved problems and/or rarely-solved instances leave some room for variation. For instance, we could define three possible properties of a problem R :

- P_1 : There is a computable instance C such that only a small class of X 's compute a solution to C .
- P_2 : For a co-small class of Z 's, Z computes an instance C_Z for which $X \oplus Z$ computes a solution to C_Z for only a small class of X 's.
- P_3 : There is an instance C such that only a small class of X 's compute a solution to C .

Phrased in terms of our definitions from above, we recognize each as speaking about rarely-solved instances; R has P_1 if there is a computable rarely-solved instance, P_2 if most Z 's compute a Z -rarely-solved instance, and P_3 if there is any rarely-solved instance at all.

Proposition 3.1. $P_1 \Rightarrow P_2 \Rightarrow P_3$, and each of these implications is strict, as witnessed by Π_2^1 -problems.

Proof. The implications are clear from the observations above. To see that P_2 does not imply P_1 , consider the principle R stating “for all X , if X is not computable, then X' exists.” Taking small to mean null, R has property P_2 but not property P_1 . Similarly for P_2 versus P_3 , consider the principle R stating “for all X , if $X = 0'$, then X' exists.” Again taking small to mean null, R has property P_3 but not property P_2 . $\square$

As it turns out, most natural problems studied in reverse mathematics satisfying property P_3 also satisfy P_1 , that is, the rarely-solved instance is witnessed by a computable instance. On its face, property P_2 seems to have few advantages, and the property itself can be argued to be less natural in this presentation. However, it has one significant advantage: under common assumptions, we can reformulate P_2 to apply to non- Π_2^1 second-order principles while maintaining the resulting ω -model separation. For this purpose, given sets $\{X_i\}_{i<\omega}$, we write $\mathcal{I}(\{X_i\}_{i<\omega})$ for the Turing ideal they generate, and (by an abuse of notation) also for the ω -model with this ideal as its second-order part.

Definition 3.2. A statement R has the *no-typical-model* (NTM) property if the class of $X = \bigoplus_{i<\omega} X_i$ such that $\mathcal{I}(\{X_i\}_{i<\omega}) \models R$ is small.

We obtain our main separation as applied to the NTM property:

Observation 3.3. If P does not have the NTM property, but Q does, then $Q \not\leq_\omega P$. In particular, P does not imply Q over RCA_0 .

Indeed, letting $\mathcal{A}$ and $\mathcal{B}$ be the classes of all $X = \bigoplus_{i<\omega} X_i$ such that $\mathcal{I}(\{X_i\}_{i<\omega}) \models P$ and $\mathcal{I}(\{X_i\}_{i<\omega}) \models Q$, respectively, $\mathcal{B}$ is small but not $\mathcal{A}$, so by Property (2) of smallness, $\mathcal{A} \not\subseteq \mathcal{B}$. In particular, there is some $X = \bigoplus_{i<\omega} X_i \in \mathcal{A} \setminus \mathcal{B}$. By definition of $\mathcal{A}$ and $\mathcal{B}$, $\mathcal{I}(\{X_i\}_{i<\omega}) \models P$ but $\mathcal{I}(\{X_i\}_{i<\omega}) \not\models Q$.

Under minor assumptions on our notion of smallness, we can show that the NTM property coincides with our property P_2 .

Definition 3.4. A *tailclass* is a class $\mathcal{C} \subseteq 2^\omega$ such that if $X \in \mathcal{C}$ and $Y =^* X$, then $Y \in \mathcal{C}$, where $=^*$ means “equals up to finite changes”. A notion of smallness is *tail-trivial* if every Borel tailclass is either small or co-small.

We should note that such 0-1 laws exist for both measure and category. For measure, it is a standard computability-theoretic application of Kolmogorov’s 0-1 Law, while for category, this is a classical result of descriptive set theory (often called a topological 0-1 law) as applied to Cantor space. Note that all our applications of tail-triviality in this article will be on particular tailclasses which are closed under Turing equivalence : if $X \in \mathcal{C}$ and $X \equiv_T Y$, then $Y \in \mathcal{C}$.

Definition 3.5. A notion of smallness is *permutation invariant* if it is closed under permuting bits. That is, if $f : \omega \rightarrow \omega$ is a permutation and $\mathcal{S}$ is small, then $\{X \circ f : X \in \mathcal{S}\}$ is also small. A notion of smallness is *computably permutation invariant* if it is closed under computable permutations.

Measure and category both yield permutation invariant notions of smallness. The main use of permutation invariance is that it lets us rearrange the columns of the sets in a small (co-small) set and retain smallness (co-smallness). For example, if $\mathcal{S}$ is small for a computably permutation invariant notion of smallness, then $\{Y \oplus X : X \oplus Y \in \mathcal{S}\}$ is also small. Permutation invariant notions of smallness also let us apply property (4) of smallness to partitions other than the one given by the usual join operation.

Theorem 3.6. Assume that our notion of smallness is tail-trivial and computably permutation invariant. A Π_2^1 -problem has the NTM property if and only if it has property P_2 .

Proof. First suppose that P has property P_2 . Let $\mathcal{S}$ be the class of all $X \oplus Y$ such that X computes a P -instance to which $X \oplus Y$ does not compute a solution. Notice that $\mathcal{S}$ is a tailclass, and it is Borel because P is a Π_2^1 -problem. Property P_2 says that $\mathcal{B}(\mathcal{S}) = \{X : \{Y : X \oplus Y \in \mathcal{S}\} \text{ is not small}\}$ is co-small. Therefore $\mathcal{S}$ is not small by property (4) of smallness and so is co-small by tail-triviality. Rearrange columns so that sets of the form $X \oplus Y$ become sets of the form $Z = \bigoplus_{i < \omega} Z_i$, where X corresponds to Z_0 and Y corresponds to $\bigoplus_{i > 0} Z_i$. The resulting class $\mathcal{R}$ of all $Z = \bigoplus_{i < \omega} Z_i$ such that Z_0 computes a P -instance to which Z does not compute a solution is co-small by computable invariance. If $Z = \bigoplus_{i < \omega} Z_i$ is in $\mathcal{R}$, then $\mathcal{I}(\{Z_i\}_{i < \omega}) \not\models P$, so P has the NTM property.

Now suppose that P does not have property P_2 . Let $\mathcal{T}$ be the class of all $X \oplus Y$ such that every X -computable P -instance has an $(X \oplus Y)$ -computable solution. The failure of property P_2 means that $\mathcal{B}(\mathcal{T}) = \{X : \{Y : X \oplus Y \in \mathcal{T}\} \text{ is not small}\}$ is not small. Thus $\mathcal{T}$ is not small by property (4) of smallness. Now let $\mathcal{R} = \{(X \oplus Y) \oplus W : X \oplus Y \in \mathcal{T}\}$. Clearly

$$\mathcal{B}(\mathcal{R}) = \{X \oplus Y : \{W : (X \oplus Y) \oplus W \in \mathcal{R}\} \text{ is not small}\} = \mathcal{T}$$

is not small, so again $\mathcal{R}$ is not small by property (4) of smallness. The classes $\mathcal{T}$ and $\mathcal{R}$ are tailclasses, and they are Borel because P is a Π_2^1 -problem. Thus they are both co-small by tail-triviality. The class $\mathcal{R}$ consists of all $(X \oplus Y) \oplus W$ where every X -computable P -instance has an $(X \oplus Y)$ -computable solution. Given n , rearrange columns so that sets of the form $(X \oplus Y) \oplus W$ become sets of the form $Z = \bigoplus_{i < \omega} Z_i$, where X corresponds to $\bigoplus_{i \leq n} Z_i$, Y corresponds to Z_{n+1} , and W corresponds to $\bigoplus_{i > n+1} Z_i$. Then the class $\mathcal{R}_n$ of all $Z = \bigoplus_{i < \omega} Z_i$ where every $(\bigoplus_{i \leq n} Z_i)$ -computable P -instance has a $(\bigoplus_{i \leq n+1} Z_i)$ -computable solution is co-small by computable invariance. The class $\bigcap_{n < \omega} \mathcal{R}_n$ is therefore co-small, and $\mathcal{I}(\{Z_i\}_{i < \omega}) \models P$ for every Z in the class. Thus P does not have the NTM property. $\square$

When showing that property P_2 implies the NTM property in the first part of the proof of Theorem 3.6, the computable invariance assumption is a matter of convenience as it allows us to be agnostic about the encoding of $\bigoplus_{i < \omega} Z_i$. The assumption may be dropped if one commits to the encoding $\bigoplus_{i < \omega} Z_i = Z_0 \oplus (\bigoplus_{i > 0} Z_i) = Z_0 \oplus (Z_1 \oplus (Z_2 \oplus \dots))$. Conversely, the computable invariance assumption seems essential to our method for showing that if property P_2 fails, then so does the NTM property.

In terms of properties P_1, P_2, P_3 , our Theorem 2.5 says that no statement with property P_3 can ω -reduce to a statement without property P_3 . Theorem 3.6 implies that Theorem 2.5 also applies to property P_2 provided that the problems in question are Π_2^1 and the notion of smallness is tail-trivial and computably permutation invariant.

Corollary 3.7. *Assume that the notion of smallness is tail-trivial and computably permutation invariant. Let P and Q be Π_2^1 -problems. If Q has property P_2 but P does not, then $Q \not\leq_\omega P$.*

Proof. In this situation, Property P_2 is equivalent to the NTM property by Theorem 3.6. Thus $Q \not\leq_\omega P$ by Observation 3.3. $\square$

Again, Theorem 2.5 implies that property P_3 is closed upward under ω -reducibility: if $Q \leq_\omega P$ and Q has property P_3 , then so does P . Likewise, Corollary 3.7 implies

that property P_2 is also closed upward under ω -reducibility, so long as the notion of smallness is tail-trivial and computably permutation invariant and the problems are Π_2^1 . The situation for property P_1 is slightly different because it only concerns computable instances.

Proposition 3.8. *If P does not have property P_1 , but Q does, then $Q \not\leq_c P$.*

Proof. Let X be the computable Q -instance such that the class $\mathcal{C}$ of sets computing a Q -solution is small. Let $\hat{X}$ be an X -computable P -instance. In particular, $\hat{X}$ is computable. Since P does not have property P_1 , the class $\mathcal{D}$ of sets computing a P -solution to $\hat{X}$ is not small. By axiom (2) of smallness, $\mathcal{D} \not\subseteq \mathcal{C}$, so there is a P -solution $\hat{Y}$ to $\hat{X}$ such that $\hat{Y}$ does not compute a Q -solution to X . In particular, $\hat{Y} \oplus X$ does not compute a Q -solution to X , so $Q \not\leq_c P$. $\square$

Many principles studied in reverse mathematics have a computability-theoretic counterpart, using the notion of *bounding degree*.

Definition 3.9. Let P be a problem. A set A is *P -bounding* over B (written $A \gg_P B$) if for every B -computable P -instance X , there is an $A \oplus B$ -computable P -solution Y .

When $B = \emptyset$, we simply say A is *P -bounding*, written $A \gg_P \emptyset$.

Theorem 3.10. *Assume that our notion of smallness is tail-trivial. A Π_2^1 -problem P has property P_1 if and only if the class of A 's that are P -bounding is small.*

Proof. Suppose first P has property P_1 , and let X be the computable P -instance witnessing it, i.e., only a small class of A 's compute a solution to X . A fortiori, only a small class of A 's can compute a solution to all computable P -instances, i.e., only a small class of A 's can be P -bounding.

Suppose now that the class of P -bounding sets is small. Fix a countable enumeration $(X_e)_{e \in \omega}$ of all computable P -instances. For every $e \in \omega$, let $\mathcal{C}_e$ be the class of all sets A which do not compute P -solutions to X_e . The fact that the class of P -bounding sets is small means that the class $\bigcup_{e \in \omega} \mathcal{C}_e$ is co-small. Since smallness is closed under arbitrary countable unions, there is some $e \in \omega$ such that $\mathcal{C}_e$ is not small. Note that $\mathcal{C}_e$ is a tailclass. Moreover, $\mathcal{C}_e$ is Borel since P is a Π_2^1 -problem, so $\mathcal{C}_e$ is co-small. In other words, the P -instance X_e witnesses that P has property P_1 . $\square$

4. APPLICATIONS TO RANDOMNESS

The goal of this section is to classify the reverse mathematics zoo in terms of which problems have property P_i for measure, for $i \in \{1, 2, 3\}$. Here by ‘measure’ we mean the Lebesgue measure (a.k.a uniform measure) on 2^ω and we denote it by μ . It can be defined as being the unique Borel probability measure such that for every $\sigma \in 2^{<\omega}$, $\mu\{X : X \upharpoonright |\sigma| = \sigma\} = 2^{-|\sigma|}$.

As we shall see, most of the statements have property P_1 (and therefore properties P_2 and P_3 by Proposition 3.1). By Proposition 3.8, it suffices to consider the weakest principles in the zoo having property P_1 and computable-reducibility. All the known principles from the zoo are known to either follow from randomness, in which case they do not even have property P_3 , or are above one of the following principles for computable reducibility:

- A Ramsey-type graph coloring principle called RCOLOR_2 (Section 4.1)

- The Atomic Model Theorem, AMT (Section 4.2)
- Cohesiveness principles, COH and CADS (Section 4.3)
- The positive measure domination principle, POS (Section 4.4)

In the remainder of this section, we shall prove that each of the principles above has property P_1 .

4.1. Ramsey-type graph coloring principles. Flood [10] introduced the *Ramsey-type weak König's lemma* (RWKL), a restriction of weak König's lemma where a solution consists of infinitely many bits of information about a path. The approach was then systematized by Bienvenu, Patey and Shafer [3] by considering Ramsey-type versions of multiple principles related to weak König's lemma. The weakest of these statements is a Ramsey-type version of a coloring principle about graphs:

Definition 4.1. Let $k \in \omega$ and $G = (V, E)$ be a graph.

- A k -coloring is a function $f : V \rightarrow k$ such that for every $x, y \in V$, if $(x, y) \in E$ then $f(x) \neq f(y)$.
- G is k -colorable if it has a k -coloring, and G is *locally k -colorable* if every finite subgraph is k -colorable.
- A set $H \subseteq V$ is k -homogeneous for G if every finite $V_0 \subseteq V$ induces a subgraph that is k -colorable by a coloring that colors every vertex in $V_0 \cap H$ color 0.

Let COLOR_k be the statement “every locally k -colorable graph is k -colorable”. Hirst [15] proved that COLOR_k is equivalent to WKL_0 over RCA_0 for every $k \geq 2$. The following statement is the Ramsey-type version of COLOR_k :

Statement 4.2. RCOLOR_k is the statement “for every infinite, locally k -colorable graph $G = (V, E)$, there is an infinite set $H \subseteq V$ that is k -homogeneous for G .”

Bienvenu, Patey and Shafer [3] proved that RCOLOR_k is equivalent to RWKL over RCA_0 for every $k \geq 3$. On the other hand, it is still unknown whether RCOLOR_2 is strictly weaker. They proved [3, Theorem 6.10] in particular that RCOLOR_2 has the P_1 property for measure:

Theorem 4.3 (Bienvenu, Patey and Shafer [3]). *There is a computable bipartite graph $G = (\omega, E)$ such that the measure of the set of oracles that enumerate homogeneous sets for G is 0.*

4.2. Atomic model theorem. The atomic model theorem is a classical statement of model theory, stating that a theory is atomic if and only if it admits an atomic model. In what follows, every theory T is *deductively closed* (for every formula φ such that $T \vdash \varphi$, $\varphi \in T$), *complete* ($T \vdash \varphi$ or $T \vdash \neg\varphi$ for every sentence φ) and *consistent* (there is no formula φ such that $T \vdash \varphi$ and $T \vdash \neg\varphi$).

Definition 4.4. Let T be a theory.

- A formula $\varphi(x_1, \dots, x_n)$ is an *atom* of T if for each formula $\psi(x_1, \dots, x_n)$, either $T \vdash \varphi \rightarrow \psi$ or $T \vdash \varphi \rightarrow \neg\psi$, but not both.
- The theory T is *atomic* if for every formula $\psi(x_1, \dots, x_n)$ consistent with T , there is an atom $\varphi(x_1, \dots, x_n)$ of T such that $T \vdash \varphi \rightarrow \psi$.
- A model $\mathcal{M}$ of T is *atomic* if every n -tuple from $\mathcal{M}$ satisfies an atom of T .

Statement 4.5. AMT is the statement “If a theory is atomic, then it admits an atomic model.”

The atomic model theorem was studied from a reverse-mathematical viewpoint by Hirschfeldt, Shore and Slaman [14], who proved that it follows from the ascending/descending sequence principle restricted to linear orders of type $\omega + \omega^*$ (SADS). Conidis [6] studied degrees bounding various model-theoretic properties, and proved in particular the following theorem. We say that a function $f : \omega \rightarrow \omega$ *dominates* $g : \omega \rightarrow \omega$ (written $f \geq g$) if $f(x) \geq g(x)$ for all but finitely many $x \in \omega$.

Theorem 4.6 (Conidis [6, Theorem 3.9]). *For every Δ_2^0 function $f : \omega \rightarrow \omega$, there is a computable atomic theory T such that every atomic model $\mathcal{M}$ of T computes a function $g : \omega \rightarrow \omega$ not dominated by f .*

Note that the original formulation is slightly different, as Conidis introduced the *isolated path property* corresponding to degrees bounding the Atomic Model Theorem, and the *escape property* corresponding to degrees relative to which $\emptyset'$ is non-high.

Dobrinen and Simpson [8] introduced the following notions of almost-everywhere domination.

Definition 4.7.

- A set A is *almost everywhere dominating* (a.e.d) if

$$\mu\{X : \forall g \leq_T X \exists f \leq_T A, f \geq g\} = 1$$

- A set A is *uniformly almost everywhere dominating* (u.a.e.d) if there is a function $f \leq_T A$ such that

$$\mu\{X : \forall g \leq_T X, f \geq g\} = 1$$

Building up on the work of Binns et al [4] and Kjos-Hanssen [20], Kjos-Hanssen, Miller and Solomon [21] proved that the two notions coincide, answering positively a conjecture of Dobrinen and Simpson [8]. Formulated in this framework, Kurtz [22] proved that $\emptyset'$ is uniformly almost everywhere dominating, from which we deduce the following theorem:

Theorem 4.8. *AMT has property P_1 for measure.*

Proof. Let $f : \omega \rightarrow \omega$ be a Δ_2^0 function witnessing that $\emptyset'$ is uniformly almost everywhere dominating. By Theorem 4.6, there is a computable atomic theory T such that every atomic model $\mathcal{M}$ of T computes a function $g : \omega \rightarrow \omega$ not dominated by f . Then, the measure of oracles computing an atomic model of T is 0. $\square$

4.3. Cohesiveness principles. Cohesiveness is closely related to the notion of *maximal set*, which was introduced by Friedberg [11] in the search for a c.e. set of incomplete degree.

Definition 4.9. An infinite set C is *cohesive* for a sequence of sets $R_0, R_1, \dots$ if for every $n \in \omega$, either $C \subseteq^* R_n$, or $C \subseteq^* \bar{R}_n$. Here, $\subseteq^*$ denotes inclusion up to finitely many elements.

Jockusch and Stephan [16] studied cohesive sets for the sequences of c.e., computable and primitive recursive sets. Cholak, Jockusch and Slaman [5] introduced the cohesiveness principle as a statement in reverse mathematics in order to divide the study of Ramsey's theorem for pairs into its stable and cohesive parts.

Statement 4.10. COH is the statement “Every countable sequence of sets admits an infinite cohesive set.”

Jockusch and Stephan [16] proved that the sequence of primitive recursive sets forms a universal instance of COH and characterized the degrees of its cohesive sets:

Theorem 4.11 (Jockusch and Stephan [16]).

- For every infinite cohesive set C for the sequence of primitive recursive sets, C' is of PA degree over $\emptyset'$.
- Let $\vec{R} = R_0, R_1, \dots$ be a uniformly computable sequence of sets. Every degree whose jump is of PA degree over $\emptyset'$ computes an infinite cohesive set for $\vec{R}$.

The relation between PA degrees and measure was extensively studied. In particular, Kučera [23] proved the following theorem:

Theorem 4.12 (Kučera [23]). Fix a set Z . Then

$$\mu\{X : X \oplus Z \text{ is of PA degree over } Z\} = 0$$

Letting $Z = \emptyset'$, this shows that the class of X such that $X \oplus \emptyset'$ is of PA degree over $\emptyset'$ is null. This is however not sufficient since one needs to consider X' instead of $X \oplus \emptyset'$. Thankfully, Kautz [19] proved the following theorem:

Theorem 4.13 (Kautz [19]). For every $n \geq 1$,

$$\mu\{X : X^{(n)} \equiv_T X \oplus \emptyset^{(n)}\} = 1$$

This enables us to classify COH with respect to measure:

Theorem 4.14. COH has property P_1 for measure.

Proof. Let $\vec{R} = R_0, R_1, \dots$ be the sequence of all primitive recursive sets. By Jockusch and Stephan (Theorem 4.11), the jump of any infinite cohesive set for $\vec{R}$ is of PA degree over $\emptyset'$. By Kautz (Theorem 4.13) and Kučera (Theorem 4.12),

$$\mu\{X : X' \text{ is of PA degree over } \emptyset'\} = \mu\{X : X \oplus \emptyset' \text{ is of PA degree over } \emptyset'\} = 0$$

It follows that the measure of sets which compute an infinite cohesive set for $\vec{R}$ is 0. $\square$

In their reverse-mathematical study of partial orders and linear orders, Hirschfeldt and Shore [13] introduced a cohesive counterpart of the Ascending Descending Sequence principle:

Statement 4.15. CADS is the statement “Every infinite linear order has a suborder that is either of type ω or ω^* or $\omega + \omega^*$.”

Hirschfeldt and Shore [13] proved in particular that COH implies CADS over RCA_0 and that the converse holds over $\text{RCA}_0 + \text{B}\Sigma_2^0$, but left the equivalence over RCA_0 open. The proofs show that COH and CADS seen as problems are computably equivalent:

Theorem 4.16 (Hirschfeldt and Shore [13]). $\text{COH} \equiv_c \text{CADS}$.

Thanks to Proposition 3.8, we deduce that

Theorem 4.17. CADS has property P_1 for measure.

4.4. Positive measure domination. The positive measure domination principle was introduced by Kjos-Hanssen, Miller and Solomon [21] in their study of almost everywhere domination.

Statement 4.18. POS is the statement “For every G_δ class $\mathcal{C} \subseteq 2^\omega$ of positive measure, there is a closed sub-class $\mathcal{D} \subseteq \mathcal{C}$ of positive measure.”

Recall the notion of almost everywhere domination (Definition 4.7). Kjos-Hanssen, Miller and Solomon [21] proved the following equivalence:

Theorem 4.19 (Kjos-Hanssen, Miller and Solomon [21]). *Let A be a set. The following are equivalent:*

- (1) *A is uniformly almost everywhere dominating*
- (2) *For every Π_2^0 class $\mathcal{C} \subseteq 2^\omega$ of positive measure, there is a $\Pi_1^0(A)$ class $\mathcal{D} \subseteq \mathcal{C}$ of positive measure.*

Any set A satisfying (2) is called *positive measure dominating*. Note that this is the computability-theoretic counterpart of the POS principle. By Martin [24], every uniformly almost everywhere dominating degree is high, and therefore, by Kautz [19] (see Theorem 4.13), the measure of sets A satisfying (2) is 0. We have all the necessary ingredients to prove the following theorem:

Theorem 4.20. POS has property P_1 for measure.

Proof. By Theorem 3.10, it suffices to show that the class of sets satisfying Theorem 4.19(2), or equivalently satisfying Theorem 4.19(1), has measure 0. By Kautz [19] (see Theorem 4.13), the measure of the class of sets of high degree is 0. By Martin’s domination theorem [24], any set satisfying Theorem 4.19(1) is of high degree, so we conclude. $\square$

5. APPLICATIONS TO GENERICITY

We now turn to the classification of the reverse mathematics zoo in terms of properties for genericity. As for randomness, many very weak principles have property P_1 for genericity, but the exact picture is slightly different. In particular, the atomic model theorem (AMT) has property P_1 for randomness, but it does not even have property P_3 for genericity.

We will give in Section 5.2 a general criterion for deciding properties for genericity over a family of Π_2^1 -problems called *restricted Π_2^1 -problems* satisfied by most principles studied in reverse mathematics.

5.1. Atomic model theorem. Recall the atomic model theorem from Section 4.2, stating that every atomic theory admits an atomic model. Hirschfeldt, Shore and Slaman [14] introduced a genericity statement closely related to the standard Henkin constructions of models of atomic theories: We say that a set $G \in 2^\omega$ *meets* a set $D \subseteq 2^{<\mathbb{N}}$ if $G \upharpoonright_n \in D$ for some $n \in \omega$.

Statement 5.1. $\Pi_1^0 G$ is the statement “For every uniformly Π_1^0 collection of sets $(D_n)_{n \in \omega}$, each of which is dense in $2^{<\mathbb{N}}$, there is a set that meets each of them.”

Hirschfeldt, Shore and Slaman [14] and Conidis [6] proved that AMT and $\Pi_1^0 G$ are computably equivalent, and that the equivalence holds over $\text{RCA}_0 + \text{IS}_2^0$. More precisely, $\text{RCA}_0 \vdash \Pi_1^0 G \rightarrow \text{AMT}$, but the converse does not hold.

Theorem 5.2 (Hirschfeldt, Shore and Slaman [14] and Conidis [6]). $\text{AMT} \equiv_c \Pi_1^0\text{G}$.

Clearly, $\Pi_1^0\text{G}$ does not have property P_3 for genericity, so neither does AMT by Theorem 2.5 and Theorem 5.2. This has to be put in contrast with measure, for which we proved that AMT has property P_1 . Because of this difference, we will need to consider problems computably above AMT to classify which admit property P_1 for genericity. Thankfully, the next section gives a general framework for deciding such a property.

5.2. Restricted Π_2^1 -problems. The following class of restricted Π_2^1 sentences was introduced by Hirschfeldt, Shore and Slaman [14] in the study of Π_1^1 -conservation theorems for AMT.

Definition 5.3. A *restricted Π_2^1 -problem* ($r\Pi_2^1$ -problem) is a problem of the form

$$\forall X[\Phi(X) \rightarrow \exists Y\Psi(X, Y)]$$

where Φ is arithmetic and Ψ is Σ_3^0 .

Note that Theorem 5.5 below will remain true even if we allow Y to be quantified over ω^ω . Many problems considered in reverse mathematics are restricted Π_2^1 . We shall actually see that there is a simple criterion to decide whether an $r\Pi_2^1$ -problem has one of those properties.

Jockusch and Soare [18, Corollary 5.2] proved the following theorem:

Theorem 5.4 (Jockusch and Soare [18]). *Let $\mathcal{C} \subseteq \omega^\omega$ be a Σ_3^0 class with no computable member. Then the class of all sets computing any member of $\mathcal{C}$ is meager.*

Note that for any computable instance X of an $r\Pi_2^1$ -problem P , the class $P(X)$ of its solutions is Σ_3^0 in the Cantor space. We therefore deduce our general criterion:

Theorem 5.5. *Let P be a $r\Pi_2^1$ -problem.*

- (1) *P has property P_1 for genericity if and only if there is a computable P -instance with no computable P -solution.*
- (2) *P has property P_2 for genericity if and only if the class of sets A computing a P -instance with no A -computable P -solution, is co-meager.*
- (3) *P has property P_3 for genericity if and only if there is a P -instance X with no X -computable P -solution.*

Proof. The forward direction for each of (1), (2) and (3) is immediate. We prove the backward directions.

- (1) Suppose that there is a computable P -instance X with no computable P -solution. The class $P(X)$ of its solutions being Σ_3^0 with no computable member, by Theorem 5.4, the class of all sets computing any member of $P(X)$ is meager, so X witnesses that P has property P_1 .
- (2) Suppose that the class of sets A computing a P -instance with no A -computable P -solutions is co-meager. For any such set A , letting X be the A -computable P -instance with no A -computable P -solution, the class $P(X)$ of its P -solutions is $\Sigma_3^0(A)$ with no A -computable member, so by Theorem 5.4 relativized to A , the class of all sets A -computing any member of $P(X)$ is meager. It follows that P has property P_2 .
- (3) Suppose that there is a P -instance X with no X -computable P -solution. The class $P(X)$ of its solutions being $\Sigma_3^0(X)$ with no X -computable member,

by Theorem 5.4 relativized to X , the class of all sets X -computing any member of $P(X)$ is meager, so X witnesses that P has property P_3 . $\square$

We now list some $\text{r}\Pi_2^1$ -problems which are minimal for property P_1 for genericity.

In their study of partial and linear orders, Hirschfeldt and Shore [13] decomposed the Ascending Descending Sequence principle (ADS) into a cohesive (CADS) and a stable (SADS) version. The CADS principle was already introduced and studied in Section 4.3 for randomness.

Statement 5.6. SADS is the statement “For every infinite linear order of type $\omega + \omega^*$, there is an infinite ascending or descending sequence.”

Tennenbaum (see Rosenstein [25]) and Denisov (see Goncharov and Nurtazin [12]) constructed a computable linear order of order type $\omega + \omega^*$ with no infinite computable ascending or descending sequence. Moreover, note that SADS is an $\text{r}\Pi_2^1$ -problem, so by Theorem 5.5(1), SADS has property P_1 for genericity.

A function $f : \omega \rightarrow \omega$ is *diagonally non- X -computable* (X -DNC) if for every $e \in \omega$, $f(e) \neq \Phi_e^X(e)$. The following statement is the reverse-mathematical counterpart of the notion of diagonally non-computable function.

Statement 5.7. DNC is the statement “For every set X , there is an X -DNC function.”

As its name suggests, an immediate diagonal argument gives that there is no computable $\emptyset$ -DNC function. Moreover, for every X , the class of X -DNC functions is either Π_1^0 in the Baire space, or Π_2^0 in the Cantor space, so DNC is an $\text{r}\Pi_2^1$ -problem. By Theorem 5.5(1), DNC has property P_1 for genericity.

Last, recall the Ramsey-type graph coloring principle RCOLOR_2 defined in Section 4.1. Given a computable, infinite locally 2-colorable graph $G = (V, E)$, the set of infinite homogeneous sets for G is a Π_1^0 class in the Baire space or a Π_2^0 class in the Cantor space, so RCOLOR_2 is an $\text{r}\Pi_2^1$ -problem. Moreover, Bienvenu, Patey and Shafer [3] constructed a computable instance of RCOLOR_2 with no computable solution. Again, by Theorem 5.5(1), RCOLOR_2 has property P_1 for genericity.

5.3. Cohesiveness principles. The analysis of cohesiveness principles for genericity is almost the same as the one for randomness from Section 4.3, except that Theorem 4.12 and Theorem 4.13 have to be replaced by their genericity counterparts:

The first theorem is the relativization of a theorem of Jockusch and Soare [18, Theorem 5.1]:

Theorem 5.8 (Jockusch and Soare [18]). *Fix a set Z . The following class is meager:*

$$\{X : X \oplus Z \text{ is of PA degree over } Z\}$$

The following theorem is considered folklore (see Jockusch [17, Lemma 2.6]), as it is at the heart of forcing:

Theorem 5.9. *For every $n \geq 1$, the following class is co-meager:*

$$\{X : X^{(n)} \equiv_T X \oplus \emptyset^{(n)}\}$$

We are now ready to classify COH and CADS:

Theorem 5.10. *COH and CADS have property P_1 for genericity.*

Proof. Let $\vec{R} = R_0, R_1, \dots$ be the sequence of all primitive recursive sets. By Jockusch and Stephan (Theorem 4.11), the jump of any infinite cohesive set for $\vec{R}$ is of PA degree over $\emptyset'$. Let $\mathcal{C} = \{X : X' \text{ is of PA degree over } \emptyset'\}$, $\mathcal{D} = \{X : X \oplus \emptyset' \text{ is of PA degree over } \emptyset'\}$ and $\mathcal{E} = \{X : X' \equiv_T X \oplus \emptyset'\}$. Note that $\mathcal{C} \subseteq (\mathcal{D} \cap \mathcal{E}) \cup \mathcal{E}^c$.

By Theorem 5.9 and Jockusch and Soare (Theorem 5.8), the classes $\mathcal{D}$ and $\mathcal{E}^c$ are meager, so $\mathcal{C}$ is meager. It follows that the class of sets which compute an infinite cohesive set for $\vec{R}$ is meager, so COH has property P_1 for genericity.

Last, by Hirschfeldt and Shore [13] (Theorem 4.16), $\text{COH} \equiv_c \text{CADS}$, so by Proposition 3.8, CADS has property P_1 for genericity. $\square$

5.4. Positive measure domination. As for cohesiveness, the analysis of the positive measure domination principle (POS) for genericity is very similar to that for randomness from Section 4.4. Indeed, the only measure-specific component of the proof was the fact that being of high degree is a measure-0 property. This also holds for genericity by Theorem 5.9.

Theorem 5.11. *POS has property P_1 for genericity.*

Proof. By Theorem 3.10, it suffices to show that the class of sets satisfying Theorem 4.19(2), or equivalently satisfying Theorem 4.19(1), is meager. By Theorem 5.9 the class of sets of high degree is meager. By Martin's domination theorem [24], any set satisfying Theorem 4.19(1) is of high degree, so we conclude. $\square$

6. SUMMARY DIAGRAMS

Based on the reverse-mathematical zoo by Astor and Dzhafarov [2], the authors extracted a zoo of Π_2^1 -problems under computable reducibility [1], and classified the problems based on which admits which property for randomness and genericity. As it happens, for both notions of smallness, every problem in the zoo either has property P_1 (in which case it also has property P_2 and P_3), or it does not even have property P_3 . On the other hand, by Proposition 3.1, one can construct artificial Π_2^1 -problems having property P_{i+1} but not P_i for $i \in \{1, 2\}$.

Open Question 6.1. Are there natural Π_2^1 -problems having property P_2 but not property P_1 , or property P_3 but not property P_2 for randomness? for genericity?

The notion of natural Π_2^1 -problem is not formally defined, but one could for instance consider any Π_2^1 -sentence appearing in a mathematical textbook at an undergraduate level.

We conclude with summary diagrams. Figure 1 and Figure 2 represent the lower part of the reverse-mathematical zoo under computable reducibility. Double arrows denote strict reductions. Problems in the hatched zone do not have property P_3 for the considered notion of typicality. Problems outside the hatched zone have property P_1 . Every other problem studied in reverse mathematics is computably above one of the problems in the diagram having property P_1 .

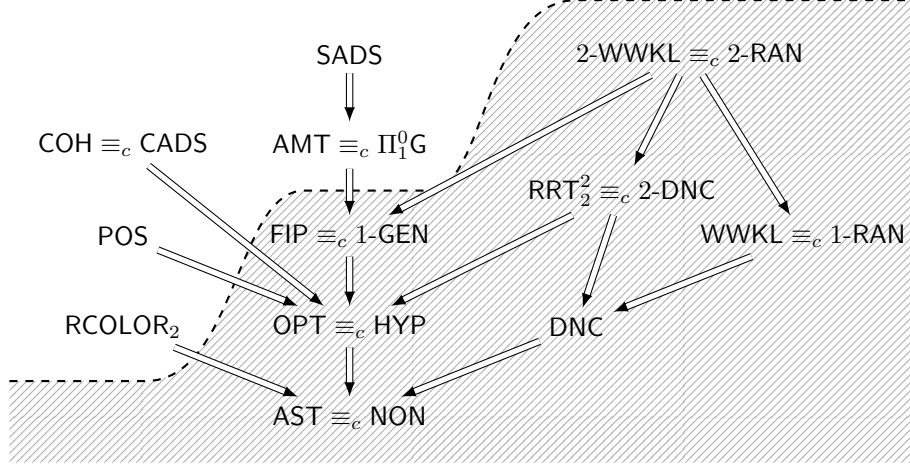


FIGURE 1. Classification based on randomness.

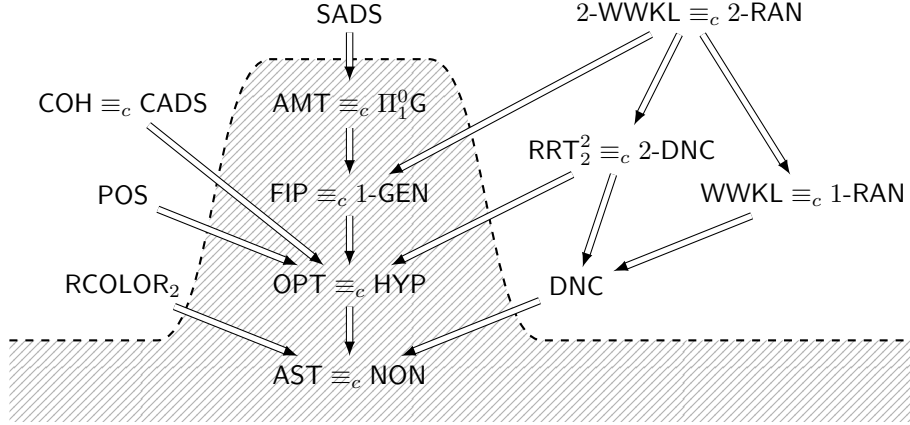


FIGURE 2. Classification based on genericity.

USE OF AI

The authors used the large language model Claude Opus 5, by Anthropic, to translate the reverse-mathematical zoo by Astor and Dzhafarov [2] into a zoo under computable reducibility [1]. Besides the dataset, the article did not use AI.

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